🚀 Data Science Roadmap

Beginner → Data Analyst → Data Scientist → AI Professional
Python-SQL-Excel-Statistics-Power BI-Machine Learning AI

📊 Phase 1: Programming & Data Fundamentals

Computer Basics
Understanding how computers store, process, and transfer data.

Why learn it?
Data Scientists work with large datasets and computing systems.

Topics to Learn:

1. Hardware vs Software
2. Operating Systems (Windows/Linux)
3. Files & Folders
4. CPU, RAM & Storage
5. Basic Networking Concepts
6. Cloud Computing Basics

Practice:
Explore your computer system, file management, and basic networking concepts.

Logic Building
The ability to solve problems systematically.

Why learn it?
Data Science requires analytical thinking and problem-solving.

Topics to Learn:

1. Flowcharts
2. Algorithms
3. Pseudocode
4. Pattern Problems
5. Logical Reasoning

Practice:
Solve everyday problems using step-by-step logic.

Python Programming
Python is the most popular programming language used in Data Science.

Why learn it?
Almost every Data Science tool and framework uses Python.

Topics to Learn:
1. Variables
2. Data Types
3. Operators
4. Loops
5. Functions
6. OOP Basics
7. Exception Handling
8. File Handling

Practice:
Build a Student Result analyser project using Python.

Data Structures Basics
Ways to organize and store data efficiently.

Why learn it?
Helps process large datasets and improves coding skills.

Topics to Learn:
1. Arrays
2. Lists
3. Tuples
4. Dictionaries
5. Sets
6. Stacks & Queues
7. Searching & Sorting

Practice:
Implement basic data structures and solve simple coding challenges.

Goal

Understand how data is stored and manipulated efficiently. By completing this phase, you should be able to:

1. ✅ Write Python programs
2. ✅ Understand programming logic
3. ✅ Solve beginner coding problems
4. ✅ Work with data structures
5. ✅ Prepare for data analysis

Outcome:
You will have a strong foundation in programming, problem-solving, and data handling concepts required for Data Science.

⬇

📈 Phase 2: Mathematics & Statistics

Mathematics and Statistics form the foundation of Data Science and Machine Learning.

Why learn it?
Machine learning algorithms are built using mathematical concepts and statistical methods to analyse data and make predictions.

Topics to Learn:

1. Mean
2. Median
3. Mode
4. Variance
5. Standard Deviation
6. Probability
7. Sampling
8. Hypothesis Testing
9. Linear Algebra Basics
10. Matrices
11. Vectors
12. Functions
13. Derivatives (Basics)

Practice:
Analyse survey data and calculate statistical measures such as average, variance, probability, and correlations to gain meaningful insights.

🎯 **Goal**
Understand how data behaves and how machine learning models make decisions using mathematical and statistical principles. By completing this phase, you should be able to:

1. ✅ Understand fundamental statistical concepts
2. ✅ analyse and interpret data distributions
3. ✅ Apply probability concepts to real-world problems
4. ✅ Work with basic linear algebra concepts
5. ✅ Build a strong foundation for Machine Learning

Outcome:
You will gain the mathematical and statistical knowledge required to analyse data effectively and understand the theory behind machine learning algorithms.

⬇️

🐼 Phase 3: Data Analysis

Data Analysis is the process of extracting meaningful insights from raw data using various tools and techniques.

Why learn it?
This is the core responsibility of Data Analysts and Data Scientists, helping organizations make data-driven decisions.

1. Topics to Learn:
2. NumPy
3. Arrays
4. Mathematical Operations
5. Data Manipulation
6. Pandas
7. DataFrames
8. Data Cleaning
9. Missing Values
10. Data Transformation
11. Duplicate Removal
12. Outlier Detection
13. Data Formatting
14. Exploratory Data Analysis (EDA)
15. Finding Patterns

16. Trend Analysis
17. Correlation Analysis

Practice:
Analyse sales data, clean the dataset, identify trends, and generate business insights using NumPy and Pandas.

🎯 **Goal**
Convert raw data into meaningful information that supports business decision-making. By completing this phase, you should be able to:

1. ✅ Work with NumPy and Pandas efficiently
2. ✅ Clean and pre-process datasets
3. ✅ Handle missing and inconsistent data
4. ✅ Perform Exploratory Data Analysis (EDA)
5. ✅ Extract meaningful insights from data

Outcome:
You will be able to collect, clean, transform, and analyse datasets to uncover patterns, trends, and valuable business insights.

⬇️

🗄️ Phase 4: SQL & Databases

SQL (Structured Query Language) is used to retrieve, manage, and analyse data stored in databases.

Why learn it?
Most business and application data are stored in databases, making SQL one of the most important skills for Data Analysts and Data Scientists.

Topics to Learn:

1. SELECT
2. WHERE
3. GROUP BY
4. ORDER BY
5. JOINS
6. Subqueries
7. Window Functions
8. MySQL
9. PostgreSQL
10. Database Design Basics

11. CRUD Operations

Practice:
Create an employee database, write SQL queries to retrieve information, generate reports, and analyse employee performance data.

🎯 **Goal**
Extract, manipulate, and analyse data efficiently using SQL and relational databases. By completing this phase, you should be able to:

1. ✅ Write SQL queries confidently
2. ✅ Retrieve and filter data efficiently
3. ✅ Use joins to combine multiple datasets
4. ✅ Perform data aggregation and analysis
5. ✅ Work with MySQL and PostgreSQL databases

Outcome:
You will be able to store, retrieve, and analyse data from databases, a critical skill required for Data Analysis, Business Intelligence, and Data Science roles.

⬇️

📊 Phase 5: Data Visualization

Data Visualization is the process of presenting data visually through charts, graphs, dashboards, and reports to make information easier to understand.

Why learn it?
Stakeholders and business leaders can understand trends, patterns, and insights more effectively through visual representations than raw numbers.

1. Topics to Learn:
2. Matplotlib
3. Seaborn
4. Plotly
5. Power BI
6. Tableau
7. Bar Charts
8. Pie Charts
9. Histograms
10. Heatmaps
11. Dashboards

Practice:
Build an interactive sales dashboard using Power BI or Tableau and visualize business performance metrics.

🎯 **Goal**
Communicate insights effectively through meaningful visualizations and dashboards. By completing this phase, you should be able to:

1. ☑ Create professional charts and graphs
2. ☑ Build interactive dashboards
3. ☑ Visualize trends and patterns in data
4. ☑ Present insights to stakeholders effectively
5. ☑ Use Power BI, Tableau, and Python visualization libraries

Outcome:
You will be able to transform complex datasets into clear, visually appealing reports and dashboards that support business decision-making.

⬇

🤖 Phase 6: Machine Learning

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed.

Why learn it?
Machine Learning powers recommendation systems, fraud detection, predictive analytics, and many modern AI applications used across industries.

Topics to Learn:

1. Linear Regression
2. Logistic Regression
3. Decision Trees
4. Random Forest
5. K-Means Clustering
6. Hierarchical Clustering
7. Accuracy
8. Precision
9. Recall
10. F1 Score

11. Confusion Matrix
12. Scikit-Learn
13. Model Training
14. Model Testing
15. Hyperparameter Tuning

Practice:
Build a house price prediction model using historical housing data and evaluate its performance using machine learning techniques.

🎯 **Goal**
Build predictive machine learning models that can learn from data and make accurate predictions. By completing this phase, you should be able to:

1. ✅ Understand supervised and unsupervised learning
2. ✅ Train and evaluate machine learning models
3. ✅ Select appropriate algorithms for different problems
4. ✅ Measure model performance using evaluation metrics
5. ✅ Build end-to-end machine learning solutions using Scikit-Learn

Outcome:
You will be able to develop, train, evaluate, and optimize machine learning models to solve real-world business and analytical problems.

⬇️

🧠 Phase 7: Deep Learning & Generative AI

Deep Learning is a subset of Artificial Intelligence that uses neural networks to learn complex patterns from large amounts of data. Generative AI focuses on creating new content such as text, images, audio, and code.

Why learn it?
Modern AI applications such as ChatGPT, image generators, recommendation systems, and intelligent assistants are powered by Deep Learning and Generative AI technologies.

Topics to Learn:

1. Neural Networks
2. Activation Functions
3. Backpropagation
4. TensorFlow

5. Keras
6. PyTorch
7. Image Classification
8. Object Detection Basics
9. Text Processing
10. Sentiment Analysis
11. Transformers Basics
12. Large Language Models (LLMs)
13. Prompt Engineering
14. RAG Concepts
15. AI Agents Basics

Practice:
Build an AI-powered chatbot that can answer questions, process user input, and generate meaningful responses using modern AI frameworks.

🎯 **Goal**
Understand and apply modern AI technologies used in real-world applications. By completing this phase, you should be able to:

1. ✅ Understand deep learning fundamentals
2. ✅ Build and train neural network models
3. ✅ Work with TensorFlow, Keras, and PyTorch
4. ✅ Understand NLP and Computer Vision basics
5. ✅ Explore Generative AI, LLMs, and AI Agents

Outcome:
You will gain practical knowledge of Deep Learning and Generative AI, enabling you to build intelligent applications and understand the technologies shaping the future of AI.

⬇️

☁️ Phase 8: MLOps & Deployment

MLOps (Machine Learning Operations) is the practice of deploying, managing, monitoring, and maintaining machine learning models in production environments.

Why learn it?
Building a machine learning model is only part of the process. Companies need reliable systems that can deploy models, serve predictions, and monitor performance in real-world applications.

Topics to Learn:

1. Flask
2. FastAPI
3. REST APIs
4. AWS Fundamentals
5. Azure Fundamentals
6. Google Cloud Fundamentals
7. MLflow
8. Model Versioning
9. Monitoring Basics
10. Docker
11. Containers
12. Deployment Basics

Practice:
Deploy a machine learning model online using Flask or FastAPI and make it accessible through a web interface or API.

🎯 **Goal**
Make AI and machine learning models available to real users through scalable and reliable deployment practices. By completing this phase, you should be able to:

1. ✅ Deploy machine learning models as APIs
2. ✅ Understand cloud platforms and hosting concepts
3. ✅ Use Docker containers for application deployment
4. ✅ Track and manage machine learning models
5. ✅ Monitor model performance in production environments

Outcome:
You will be able to deploy, manage, and maintain machine learning solutions in production, bridging the gap between model development and real-world usage.

⬇️

💼 Phase 9: Build Real Projects

Building real-world projects allows you to apply your knowledge, gain hands-on experience, and demonstrate your skills to recruiters and employers.

Why learn it?
Projects are one of the best ways to showcase practical experience and stand out in Data Science, Machine Learning, and AI job applications.

Topics to Learn:

1. Student Performance Predictor
2. Sales Analysis Dashboard
3. Movie Recommendation System
4. Customer Churn Prediction
5. Credit Risk Analysis
6. Stock Market Trend Analysis
7. Resume Screening System
8. AI Chatbot
9. Fraud Detection System
10. Medical Diagnosis Assistant
11. Recommendation Engine

Practice:
Build projects of increasing complexity, publish the source code on GitHub, and create a portfolio showcasing your work and achievements.

🎯 **Goal**
Build a strong portfolio that demonstrates practical Data Science, Machine Learning, and AI skills. By completing this phase, you should be able to:

1. ☑️ Apply theoretical concepts to real-world problems
2. ☑️ Develop end-to-end Data Science projects
3. ☑️ Create a professional GitHub portfolio
4. ☑️ Demonstrate problem-solving and analytical skills
5. ☑️ Showcase practical experience to recruiters and hiring managers

Outcome:
You will have a portfolio of real-world projects that validates your technical skills and significantly improves your chances of securing Data Science and AI-related roles.

⬇️

🎯 Phase 10: Job Preparation

Job preparation focuses on creating a strong professional profile, showcasing your skills, and preparing for technical and behavioural interviews.

Why learn it?
Having technical skills is important, but presenting them effectively is essential to secure interviews and land your first Data Science or AI role.

Topics to Learn:

1. One-page Resume
2. ATS Friendly Format
3. GitHub Portfolio
4. Professional LinkedIn Headline
5. Projects & Certifications
6. Python Interview Questions
7. SQL Interview Questions
8. Statistics Fundamentals
9. Machine Learning Concepts
10. Case Studies
11. Business Problem Solving
12. Data Interpretation
13. LinkedIn
14. Naukri
15. Foundit
16. NextGenZ.careers

Practice:
Create an ATS-friendly resume, optimize your LinkedIn profile, build a GitHub portfolio, and participate in mock interviews and case study discussions.

🎯 Goal

Become interview-ready and secure your first Data Science, Machine Learning, or AI role. By completing this phase, you should be able to:

1. ✅ Create a professional resume and portfolio
2. ✅ Build a strong LinkedIn and GitHub presence
3. ✅ Answer technical interview questions confidently
4. ✅ Solve business and analytical case studies
5. ✅ Apply effectively for Data Science and AI opportunities

Outcome:
You will be fully prepared to apply for and succeed in interviews for Data Analyst, Data Scientist, Machine Learning Engineer, and AI Engineer roles.

🏆 **Final Outcome**
By completing this roadmap, you will:

1. ✅ Write professional Python code
2. ✅ Analyse and visualize data
3. ✅ Work with SQL databases
4. ✅ Build machine learning models
5. ✅ Understand Deep Learning and Generative AI
6. ✅ Deploy AI solutions to production
7. ✅ Build a strong Data Science portfolio
8. ✅ Become job-ready for Data Analyst, Data Scientist, ML Engineer, and AI Engineer roles in 2026

Outcome:
You will possess the technical knowledge, practical experience, and industry-ready skills required to build a successful career in Data Science, Machine Learning, and Artificial Intelligence.